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Studierichting: Informatica
Oefeningenreeks 4A nr. 37.11

$$\int \frac{dx}{4x^2 + 12x + 18}$$
$$= \int \frac{dx}{(2x+3)^2 + 9}$$
$$= \frac{1}{9} \int \frac{1}{\frac{(2x+3)^2}{9} + 1} dx$$
$$= \frac{1}{9} \int \frac{dx}{\left(\frac{2x+3}{3}\right)^2 + 1}$$
$$t = \frac{2x+3}{3} \Rightarrow dt = \frac{2}{3} dx \Rightarrow \frac{3}{2} dt = dx$$
$$= \frac{1}{9} \cdot \frac{3}{2} \int \frac{dt}{t^2 + 1}$$
$$= \frac{1}{6} \operatorname{Bgtan}(t) + C$$
$$= \frac{1}{6} \operatorname{Bgtan}\left(\frac{2x+3}{3}\right) + C$$

References